

TEJAS NEHETE

+91 8767836571 | tejasdn2004@gmail.com | GitHub | LinkedIn

EDUCATION

Woodridge High School (Central Board of Secondary Education) Class X | 92.8% | 2021
Spring Dales Junior College (Maharashtra State Board) Class XII | 76.33% | 2023
VIT Bhopal University (B.Tech in Computer Science & Engineering) | CGPA: 8.93/10 | 2023-27

EXPERIENCE

Indian Institute of Technology Indore (IIT Indore)

Research Internship | May 2025 - June 2025

- Developed an AI-powered ICU event tracking and data collection system using computer vision, enabling real-time monitoring of medical devices and ICU activities through PTZ camera feeds.
- Fine-tuned YOLOv8 for object detection by training on sample datasets and a Brain Tumor MRI dataset, gaining expertise in annotation pipelines, data preprocessing, and medical image analysis.
- Annotated 250+ ICU images to evaluate Qwen Vision-Language Models and explored Qwen + LoRA and Grounding DINO + SAM for automated annotation and zero-shot segmentation in healthcare applications.
- Manually annotated 500 ICU images from the AIIMS dataset across multiple object classes (Ventilator, Monitor, Bed Number, Face, etc.) and implemented incremental learning with YOLOv7 to improve model adaptability for new ICU scenarios.

PROJECTS

Brain Tumor Detection Tool

Tech Stack: TensorFlow | OpenCV | HTML/CSS | JavaScript

- Created a brain tumor detection system using TensorFlow and OpenCV to classify MRI scans, improving automated medical image analysis and diagnostic support.
- Built and trained a Convolutional Neural Network (CNN) for brain tumor classification, incorporating image preprocessing, augmentation, and performance evaluation to enhance prediction accuracy.
- Pioneered an image processing pipeline using OpenCV for MRI preprocessing, including resizing, normalization, and noise reduction to improve model performance.

Task Management System

Tech Stack: Flask | MySQL | HTML | CSS | Bootstrap

- Formulated a role-based web application featuring 2 user roles (Manager and Employee) with secure authentication and authorization.
- Streamlined 5+ core modules, including task assignment, progress tracking, deadline management, employee dashboard, and status updates.
- Designed a normalized MySQL database with 6+ relational tables and optimized CRUD operations using SQL queries.

TECHNICAL SKILLS

- Programming Languages : Python • C++ • Java • JavaScript • HTML • CSS • SQL • PL/SQL
- Frameworks & Libraries : Flask • NumPy • Pandas • Scikit-learn • Matplotlib • Seaborn • Pygame • Bootstrap
- Developer Tools : Git • GitHub • VS Code • Jupyter Notebook • Excel
- Operating Systems & Linux : Ubuntu • Bash • AWK • Sed • Grep • LVM • User & Permission Management
- AI / Machine Learning : Machine Learning Pipelines • Data Preprocessing • Feature Engineering • Model Evaluation • Flask Deployment • Pickle • Joblib